

第九章多元微分学练习题（一）解答

1. 设 $f(0,0) = 0$, 当 $(x, y) \neq (0,0)$ 时, $f(x, y)$ 为如下形式之一, 则函数 $f(x, y)$ 在点 $(0,0)$ 处连续的是 ()

(A) $\frac{xy}{x^2 + y^2}$; (B) $\frac{x^2 + y^2}{x + y}$; (C) $\frac{xy^2}{x^2 + y^4}$; (D) $\frac{xy^2}{x^2 + y^2}$.

解 对于(A), 当 (x, y) 沿直线 $y = kx$ 趋于 $(0,0)$ 时, 有

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=kx}} f(x, y) = \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=kx}} \frac{xy}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x \cdot kx}{x^2 + (kx)^2} = \frac{k}{1 + k^2}$$

不同的 k 极限不相等, 所以 $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ 不存在, 故函数 $f(x, y)$ 在点 $(0,0)$ 处不连续, 排除(A).

对于(B), 当 (x, y) 沿直线 $y = 0$ 趋于 $(0,0)$ 时, 有

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=0}} f(x, y) = \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=0}} \frac{x^2 + y^2}{x + y} = \lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

而当 (x, y) 沿直线 $y = x^2 - x$ 趋于 $(0,0)$ 时, 有

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=x^2-x}} f(x, y) = \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=x^2-x}} \frac{x^2 + y^2}{x + y} = \lim_{x \rightarrow 0} \frac{x^2 + (x^2 - x)^2}{x^2} = 2$$

沿不同路径极限不相等, 所以 $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ 不存在, 故函数 $f(x, y)$ 在点 $(0,0)$ 处不连续, 排除(B).

对于(C), 当 (x, y) 沿直线 $y = 0$ 趋于 $(0,0)$ 时, 有

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=0}} f(x, y) = \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=0}} \frac{xy^2}{x^2 + y^4} = \lim_{x \rightarrow 0} 0 = 0$$

而当 (x, y) 沿直线 $x = y^2$ 趋于 $(0,0)$ 时, 有

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ x=y^2}} f(x, y) = \lim_{\substack{(x,y) \rightarrow (0,0) \\ x=y^2}} \frac{y^4}{y^4 + y^4} = \frac{1}{2}$$

沿不同路径极限不相等, 所以 $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ 不存在, 故函数 $f(x, y)$ 在点 $(0,0)$ 处不连续, 排除(C).

对于(D), 令 $x = r \cos \theta, y = r \sin \theta$, 则 $(x, y) \rightarrow (0,0) \Leftrightarrow r \rightarrow 0$, 于是

$$\begin{aligned}\lim_{(x,y)\rightarrow(0,0)} f(x,y) &= \lim_{(x,y)\rightarrow(0,0)} \frac{xy^2}{x^2+y^2} = \lim_{x\rightarrow 0} \frac{(r\cos\theta)(r\sin\theta)^2}{r^2} \\ &= \lim_{x\rightarrow 0} r\cos\theta\sin^2\theta = 0 = f(0,0)\end{aligned}$$

所以函数 $f(x, y)$ 在点 $(0,0)$ 处连续, 故选(D).

2. 已知 $f(x, y) = e^{\sqrt{x^2+y^4}}$, 则 ()

(A) $f'_x(0,0), f'_y(0,0)$ 都存在; (B) $f'_x(0,0)$ 存在, $f'_y(0,0)$ 不存在;

(C) $f'_x(0,0)$ 不存在, $f'_y(0,0)$ 存在; (D) $f'_x(0,0), f'_y(0,0)$ 都不存在.

解 由偏导数定义, 极限

$$\lim_{x\rightarrow 0} \frac{f(x,0) - f(0,0)}{x - 0} = \lim_{x\rightarrow 0} \frac{e^{|x|} - 1}{x} = \lim_{x\rightarrow 0} \frac{|x|}{x}$$

不存在, 所以 $f'_x(0,0)$ 不存在.

由偏导数定义, 有

$$f'_y(0,0) = \lim_{y\rightarrow 0} \frac{f(0,y) - f(0,0)}{y - 0} = \lim_{y\rightarrow 0} \frac{e^{y^2} - 1}{y} = \lim_{y\rightarrow 0} \frac{y^2}{y} = 0$$

所以 $f'_y(0,0)$ 存在.

因此, 选(C).

3. 二元函数 $f(x, y)$ 在点 $(0,0)$ 处可微的一个充分条件是 ()

(A) $\lim_{(x,y)\rightarrow(0,0)} [f(x,y) - f(0,0)] = 0$;

(B) $\lim_{x\rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = 0, \lim_{y\rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = 0$;

(C) $\lim_{(x,y)\rightarrow(0,0)} \frac{f(x,y) - f(0,0)}{\sqrt{x^2+y^2}} = 0$;

(D) $\lim_{x\rightarrow 0} [f'_x(x,0) - f'_x(0,0)] = 0, \lim_{y\rightarrow 0} [f'_y(0,y) - f'_y(0,0)] = 0$.

解 (A)表明 $f(x, y)$ 在点 $(0,0)$ 处连续, 但连续不能保证可微(见教材或课堂笔记中的例子),

排除(A).

由(B)可知

$$f'_y(0,0) = \lim_{y\rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = 0$$

所以 $f'_x(0,0), f'_y(0,0)$ 都存在, 但偏导数存在不能保证可微 (见教材或课堂笔记中的例子),

排除(B).

对于(C), 由于

$$\begin{aligned} & \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - 0 \cdot (x-0) - 0 \cdot (y-0)}{\sqrt{(x-0)^2 + (y-0)^2}} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{\sqrt{x^2 + y^2}} = 0 \end{aligned}$$

由可微的等价形式知, 函数 $f(x,y)$ 在点 $(0,0)$ 处可微, 且

$$dz|_{(0,0)} = 0 \cdot (x-0) + 0 \cdot (y-0) = 0$$

因此选(C).

对于(D), 有

$$\begin{aligned} & \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=0}} [f'_x(x,y) - f'_x(0,0)] = \lim_{x \rightarrow 0} [f'_x(x,0) - f'_x(0,0)] = 0 \\ & \lim_{\substack{(x,y) \rightarrow (0,0) \\ x=0}} [f'_x(x,y) - f'_x(0,0)] = \lim_{y \rightarrow 0} [f'_x(0,y) - f'_x(0,0)] = 0 \end{aligned}$$

仅表示一元函数 $f'_x(x,0)$ 在 $x=0$ 处连续和一元函数 $f'_x(0,y)$ 在 $y=0$ 处连续, 不是偏导

函数 $f'_x(x,y), f'_y(x,y)$ 在点 $(0,0)$ 处连续 (由定理知偏导函数连续必可微), 因此无法推出函数的可微性, 故排除(D)

4. 设函数 $f(x,y)$ 在点 $(0,1)$ 处连续, 则下列命题正确的是 ()

(A) 若 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x,y)}{|x| + |y-1|}$ 存在, 则 $f(x,y)$ 在点 $(0,1)$ 处可微;

(B) 若 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x,y) - 2x + y - 2}{x^2 + (y-1)^2}$ 存在, 则 $f(x,y)$ 在点 $(0,1)$ 处可微;

(C) 若 $f(x,y)$ 在点 $(0,1)$ 处可微, 则 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x,y)}{|x| + |y-1|}$ 存在;

(D) 若 $f(x,y)$ 在点 $(0,1)$ 处可微, 则 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x,y) - 2x + y - 2}{x^2 + (y-1)^2}$ 存在.

解 对于(A), 取 $f(x,y) = |x| + |y-1|$, 则 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x,y)}{|x| + |y-1|} = 1$, 但易知

$f'_x(0,1), f'_y(0,1)$ 均不存在, 因此 $f(x,y) = |x| + |y-1|$ 在点 $(0,1)$ 处不可微, 故排除(A).

对于(B), 因为

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{x^2 + (y - 1)^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{\frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y - 1)^2}}}{\sqrt{x^2 + (y - 1)^2}}$$

存在, 而当 $x \rightarrow 0, y \rightarrow 1$ 时 $\rho = \sqrt{x^2 + (y - 1)^2} \rightarrow 0$, 所以

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y - 1)^2}} = 0$$

由于 $f(x, y)$ 在点 $(0, 1)$ 处连续, 根据上式可知

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} (f(x, y) - 2x + y - 2) = f(0, 1) - 1 = 0 \Rightarrow f(0, 1) = 1$$

因此

$$\begin{aligned} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - f(0, 1) - 2 \cdot (x - 0) + 1 \cdot (y - 1)}{\sqrt{(x - 0)^2 + (y - 1)^2}} \\ \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y - 1)^2}} = 0 \end{aligned}$$

由可微的等价形式知, 函数 $f(x, y)$ 在点 $(0, 1)$ 处可微, 且微分为

$$dz|_{(0, 1)} = 2 \cdot (x - 0) - 1 \cdot (y - 1) = 2\Delta x - \Delta y = 2dx - dy$$

对于(C)和(D), 取 $f(x, y) = 1$, 则显然 $f(x, y)$ 可微, 但

$$\begin{aligned} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y)}{|x| + |y - 1|} &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{1}{|x| + |y - 1|} \\ \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{x^2 + (y - 1)^2} &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{-2x + y - 1}{x^2 + (y - 1)^2} \end{aligned}$$

均不存在, 故排除(C)和(D).

5. 设函数 $f(x, y)$ 可微, 且对任意 x, y 都有 $\frac{\partial f(x, y)}{\partial x} > 0, \frac{\partial f(x, y)}{\partial y} < 0$, 则使不等式

$f(x_1, y_1) < f(x_2, y_2)$ 成立的一个充分条件是 ()

(A) $x_1 > x_2, y_1 < y_2$; (B) $x_1 > x_2, y_1 > y_2$;

(C) $x_1 < x_2, y_1 < y_2$; (D) $x_1 < x_2, y_1 > y_2$.

解 因为

$$\begin{aligned} f(x_2, y_2) - f(x_1, y_1) &= (f(x_2, y_2) - f(x_1, y_2)) + (f(x_1, y_2) - f(x_1, y_1)) \\ &= f'_x(\xi, y_2)(x_2 - x_1) + f'_y(x_1, \eta)(y_2 - y_1) \end{aligned}$$

其中 ξ 介于 x_1 和 x_2 之间, η 介于 y_1 和 y_2 之间, 而 $f'_x(\xi, y_2) > 0, f'_y(x_1, \eta) < 0$, 所以当

$x_2 - x_1 > 0, y_2 - y_1 < 0$ 时, $f(x_2, y_2) - f(x_1, y_1) > 0$, 故选(D).

(利用拉格朗日中值公式 $f(b) - f(a) = f'(\xi)(b - a)$, 可得

$f(b, y) - f(a, y) = f'_x(\xi, y)(b - a), f(x, b) - f(x, a) = f'_y(x, \eta)(b - a)$, ξ, η 介于 a 和 b 之间)

6. 设函数 $f(u, v)$ 由关系式 $f(xg(y), y) = x + g(y)$ 确定, 其中 $g(y)$ 可微, 且

$$g'(y) \neq 0, \text{ 求 } \frac{\partial^2 f}{\partial u \partial v}.$$

解 令 $u = xg(y), v = y$, 则 $x = \frac{u}{g(v)}, y = v$, 代入关系式得

$$f(u, v) = \frac{u}{g(v)} + g(v)$$

求偏导数得

$$\frac{\partial f}{\partial u} = \frac{\partial}{\partial u} \left(\frac{u}{g(v)} + g(v) \right) = \frac{1}{g(v)}$$

$$\frac{\partial^2 f}{\partial u \partial v} = \frac{\partial}{\partial v} \left(\frac{1}{g(v)} \right) = -\frac{g'(v)}{(g(v))^2}$$

7. 设函数 $z = e^{-x} - f(x - 2y)$, 且当 $y = 0$ 时 $z = x^2$, 求 $\frac{\partial z}{\partial x}$.

解 令 $y = 0$, 得

$$z = e^{-x} - f(x) = x^2$$

解得 $f(x) = e^{-x} - x^2$, 于是 $z = e^{-x} - e^{-(x-2y)} + (x - 2y)^2$, 求偏导得

$$\frac{\partial z}{\partial x} = -e^{-x} + e^{-(x-2y)} + 2(x - 2y)$$

8. 设 $z = f(x - y, x^2 + y^2) + f(xy, x)$, 其中 f 可微, 求 $\frac{\partial z}{\partial x}$.

解 (此题不能省略中间变量, 否则意义会产生混淆, 例如, $f_1'(x-y, x^2+y^2)$ 和

$f_1'(xy, x)$ 省略中间变量后都写成 f_1' , 但显然两者意义不同)

求偏导得

$$\begin{aligned}\frac{\partial z}{\partial x} &= f_1'(x-y, x^2+y^2) \cdot \frac{\partial(x-y)}{\partial x} + f_2'(x-y, x^2+y^2) \cdot \frac{\partial(x^2+y^2)}{\partial x} \\ &+ f_1'(xy, x) \cdot \frac{\partial(xy)}{\partial x} + f_2'(xy, x) \cdot \frac{\partial x}{\partial x} \\ &= f_1'(x-y, x^2+y^2) + 2xf_2'(x-y, x^2+y^2) + yf_1'(xy, x) + f_2'(xy, x)\end{aligned}$$

9. 设函数 $z = f(x, y)$ 具有连续偏导数, 将 $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$ 表示成极坐标系下的公式.

解 方法一、作极坐标变换 $x = r \cos \theta, y = r \sin \theta$, 则

$$z = f(x, y) = f(r \cos \theta, r \sin \theta) = g(r, \theta)$$

将 r, θ 视为自变量, x, y 视为中间变量, 求偏导得

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial z}{\partial x} \cos \theta + \frac{\partial z}{\partial y} \sin \theta$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial z}{\partial x} \cdot (-r \sin \theta) + \frac{\partial z}{\partial y} (r \cos \theta)$$

(这里我们求出了两个偏导数, 但实际只用到第一个偏导数)

第一式两边同乘 r 得

$$r \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} r \cos \theta + \frac{\partial z}{\partial y} r \sin \theta$$

所以

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{\partial z}{\partial x} r \cos \theta + \frac{\partial z}{\partial y} r \sin \theta = r \frac{\partial z}{\partial r}$$

方法二、由 $x = r \cos \theta, y = r \sin \theta$ 得 $r = \sqrt{x^2 + y^2}, \theta = \arctan \frac{y}{x}$, 所以

$$z = g(r, \theta) = g\left(\sqrt{x^2 + y^2}, \arctan \frac{y}{x}\right) = f(x, y)$$

将 x, y 视为自变量, r, θ 视为中间变量, 求偏导得

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{\partial z}{\partial r} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \left(-\frac{y}{x^2 + y^2}\right)$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial z}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{\partial z}{\partial r} \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{x}{\sqrt{x^2 + y^2}}$$

于是

$$\begin{aligned} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= \frac{\partial z}{\partial r} \cdot \frac{x^2}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \left(-\frac{xy}{\sqrt{x^2 + y^2}} \right) \\ &+ \frac{\partial z}{\partial r} \cdot \frac{y^2}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{xy}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2} \frac{\partial z}{\partial r} = r \frac{\partial z}{\partial r} \end{aligned}$$

10. 设函数 $z = f(x^2 - y^2, e^{xy})$, 其中 f 具有连续的二阶偏导数, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x \partial y}$.

解 求偏导得

$$\begin{aligned} \frac{\partial z}{\partial x} &= f'_1 \cdot \frac{\partial(x^2 - y^2)}{\partial x} + f'_2 \cdot \frac{\partial(e^{xy})}{\partial x} = 2xf'_1 + ye^{xy} f'_2 \\ \frac{\partial z}{\partial y} &= f'_1 \cdot \frac{\partial(x^2 - y^2)}{\partial y} + f'_2 \cdot \frac{\partial(e^{xy})}{\partial y} = -2yf'_1 + xe^{xy} f'_2 \end{aligned}$$

求二阶偏导数得

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} (2xf'_1 + ye^{xy} f'_2) = 2x \frac{\partial}{\partial y} (f'_1) + \frac{\partial}{\partial y} (ye^{xy}) \cdot (f'_2) + ye^{xy} \frac{\partial}{\partial y} (f'_2) \\ &= 2x \left[f''_{11} \cdot \frac{\partial(x^2 - y^2)}{\partial y} + f''_{12} \cdot \frac{\partial(e^{xy})}{\partial y} \right] + (xy + 1)e^{xy} \cdot (f'_2) \\ &+ ye^{xy} \left[f''_{21} \cdot \frac{\partial(x^2 - y^2)}{\partial y} + f''_{22} \cdot \frac{\partial(e^{xy})}{\partial y} \right] \\ &= 2x [f''_{11} \cdot (-2y) + f''_{12} \cdot xe^{xy}] + (xy + 1)e^{xy} f'_2 + ye^{xy} [f''_{21} \cdot (-2y) + f''_{22} \cdot xe^{xy}] \\ &= -4xyf''_{11} + 2(x^2 - y^2)e^{xy} f''_{12} + xye^{2xy} f''_{22} + (xy + 1)e^{xy} f'_2 \end{aligned}$$

11. 设函数 $u = f(x, y, z)$, $z = -x^2 \sin \ln y$, 其中 f 具有连续的二阶偏导数, 求

$$\frac{\partial^2 u}{\partial x \partial y}.$$

解 求偏导得

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial x} \\ &= \frac{\partial f}{\partial x} \cdot 1 + \frac{\partial f}{\partial y} \cdot 0 + \frac{\partial f}{\partial z} \cdot (-2x \sin \ln y) = \frac{\partial f}{\partial x} - 2x \sin \ln y \frac{\partial f}{\partial z} \end{aligned}$$

求二阶偏导数

$$\begin{aligned}
\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} - 2x \sin \ln y \frac{\partial f}{\partial z} \right) \\
&= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) - \frac{\partial (2x \sin \ln y)}{\partial y} \frac{\partial f}{\partial z} - 2x \sin \ln y \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial z} \right) \\
&= \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial x}{\partial y} + \frac{\partial^2 f}{\partial x \partial y} \frac{\partial y}{\partial y} + \frac{\partial^2 f}{\partial x \partial z} \frac{\partial z}{\partial y} \right] - \frac{2x \cos \ln y}{y} \frac{\partial f}{\partial z} \\
&\quad - 2x \sin \ln y \left[\frac{\partial^2 f}{\partial z \partial x} \frac{\partial x}{\partial y} + \frac{\partial^2 f}{\partial z \partial y} \frac{\partial y}{\partial y} + \frac{\partial^2 f}{\partial z^2} \frac{\partial z}{\partial y} \right] \\
&= \left[\frac{\partial^2 f}{\partial x^2} \cdot 0 + \frac{\partial^2 f}{\partial x \partial y} \cdot 1 + \frac{\partial^2 f}{\partial x \partial z} \cdot \left(-\frac{x^2 \cos \ln y}{y} \right) \right] - \frac{2x \cos \ln y}{y} \frac{\partial f}{\partial z} \\
&\quad - 2x \sin \ln y \left[\frac{\partial^2 f}{\partial z \partial x} \cdot 0 + \frac{\partial^2 f}{\partial z \partial y} \cdot 1 + \frac{\partial^2 f}{\partial z^2} \cdot \left(-\frac{x^2 \cos \ln y}{y} \right) \right] \\
&= \frac{\partial^2 f}{\partial x \partial y} - \frac{x^2 \cos \ln y}{y} \frac{\partial^2 f}{\partial x \partial z} - 2x \sin \ln y \frac{\partial^2 f}{\partial z \partial y} \\
&\quad + \frac{2x^3 \sin \ln y \cos \ln y}{y} \frac{\partial^2 f}{\partial z^2} - \frac{2x \cos \ln y}{y} \frac{\partial f}{\partial z}
\end{aligned}$$

(注意, $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \neq \frac{\partial^2 f}{\partial x \partial y}$, 这是因为此不等式左边的 y 是自变量, 右边的 y 是中间变

量 (y 有两重身份, 既是自变量又是中间变量), $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ 的含义是复合函数

$\frac{\partial f}{\partial x} = f'_x(x, y, z)$, $z = -x^2 \sin \ln y$ 关于自变量 y 求偏导 (不是对中间变量 y 求偏

导!!), 同理 $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial z} \right) \neq \frac{\partial^2 f}{\partial z \partial y}$)

12. 设函数 $g(x, y) = f\left(\frac{y}{x}\right) + yf\left(\frac{x}{y}\right)$, 其中 f 具有二阶导数, 求 $x^2 \frac{\partial^2 g}{\partial x^2} - y^2 \frac{\partial^2 g}{\partial y^2}$.

解 (此题不能省略中间变量, 否则意义会产生混淆, 例如, $f\left(\frac{y}{x}\right)$ 和 $f\left(\frac{x}{y}\right)$ 省略中间变量

后都写成 f , 但显然两者意义不同)

求偏导得

$$\begin{aligned}
\frac{\partial g}{\partial x} &= f'\left(\frac{y}{x}\right) \frac{\partial\left(\frac{y}{x}\right)}{\partial x} + y f'\left(\frac{x}{y}\right) \frac{\partial\left(\frac{x}{y}\right)}{\partial x} \\
&= f'\left(\frac{y}{x}\right) \cdot \left(-\frac{y}{x^2}\right) + y f'\left(\frac{x}{y}\right) \cdot \frac{1}{y} = -\frac{y}{x^2} f'\left(\frac{y}{x}\right) + f'\left(\frac{x}{y}\right) \\
\frac{\partial g}{\partial y} &= f'\left(\frac{y}{x}\right) \frac{\partial\left(\frac{y}{x}\right)}{\partial y} + f\left(\frac{x}{y}\right) + y f'\left(\frac{x}{y}\right) \frac{\partial\left(\frac{x}{y}\right)}{\partial y} \\
&= f'\left(\frac{y}{x}\right) \cdot \frac{1}{x} + f\left(\frac{x}{y}\right) + y f'\left(\frac{x}{y}\right) \cdot \left(-\frac{x}{y^2}\right) = \frac{1}{x} f'\left(\frac{y}{x}\right) + f\left(\frac{x}{y}\right) - \frac{x}{y} f'\left(\frac{x}{y}\right)
\end{aligned}$$

求二阶偏导得

$$\begin{aligned}
\frac{\partial^2 g}{\partial x^2} &= \frac{\partial}{\partial x} \left(-\frac{y}{x^2} f'\left(\frac{y}{x}\right) + f'\left(\frac{x}{y}\right) \right) \\
&= \frac{2y}{x^3} f'\left(\frac{y}{x}\right) - \frac{y}{x^2} f''\left(\frac{y}{x}\right) \cdot \left(-\frac{y}{x^2}\right) + f''\left(\frac{x}{y}\right) \cdot \frac{1}{y} \\
&= \frac{2y}{x^3} f'\left(\frac{y}{x}\right) + \frac{y^2}{x^4} f''\left(\frac{y}{x}\right) + \frac{1}{y} f''\left(\frac{x}{y}\right) \\
\frac{\partial^2 g}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{1}{x} f'\left(\frac{y}{x}\right) + f\left(\frac{x}{y}\right) - \frac{x}{y} f'\left(\frac{x}{y}\right) \right) \\
&= \frac{1}{x} f''\left(\frac{y}{x}\right) \cdot \frac{1}{x} + f'\left(\frac{x}{y}\right) \cdot \left(-\frac{x}{y^2}\right) + \frac{x}{y^2} f'\left(\frac{x}{y}\right) - \frac{x}{y} f''\left(\frac{x}{y}\right) \cdot \left(-\frac{x}{y^2}\right) \\
&= \frac{1}{x^2} f''\left(\frac{y}{x}\right) + \frac{x^2}{y^3} f''\left(\frac{x}{y}\right)
\end{aligned}$$

因此

$$\begin{aligned}
x^2 \frac{\partial^2 g}{\partial x^2} - y^2 \frac{\partial^2 g}{\partial y^2} &= \left[\frac{2y}{x} f'\left(\frac{y}{x}\right) + \frac{y^2}{x^2} f''\left(\frac{y}{x}\right) + \frac{x^2}{y} f''\left(\frac{x}{y}\right) \right] \\
&\quad - \left[\frac{y^2}{x^2} f''\left(\frac{y}{x}\right) + \frac{x^2}{y} f''\left(\frac{x}{y}\right) \right] = \frac{2y}{x} f'\left(\frac{y}{x}\right)
\end{aligned}$$

13. 设 $z = z(x, y)$ 是由方程 $x^2 + y^2 - z = \varphi(x + y + z)$ 所确定的函数, 其中 φ 具有

连续的二阶导数, 且 $\varphi' \neq -1$, (1) 求 dz ; (2) 记 $u(x, y) = \frac{1}{x - y} \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)$, 求 $\frac{\partial u}{\partial x}$.

解 (1) 方程两边取全微分

$$d(x^2 + y^2 - z) = d[\varphi(x + y + z)]$$

经计算得

$$2xdx + 2ydy - dz = \varphi' \cdot (dx + dy + dz)$$

解出 dz 得

$$dz = \frac{2x - \varphi'}{1 + \varphi'} dx + \frac{2y - \varphi'}{1 + \varphi'} dy$$

(2) 由 (1) 中 dz 的表达式知

$$\frac{\partial z}{\partial x} = \frac{2x - \varphi'}{1 + \varphi'}, \quad \frac{\partial z}{\partial y} = \frac{2y - \varphi'}{1 + \varphi'}$$

所以

$$u(x, y) = \frac{1}{x - y} \left(\frac{2x - \varphi'}{1 + \varphi'} - \frac{2y - \varphi'}{1 + \varphi'} \right) = \frac{2}{1 + \varphi'}$$

求偏导得

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{2}{1 + \varphi'} \right) = 2 \cdot \left(-\frac{1}{(1 + \varphi')^2} \right) \cdot \frac{\partial}{\partial x} (\varphi') \\ &= 2 \cdot \left(-\frac{1}{(1 + \varphi')^2} \right) \cdot \varphi'' \cdot \frac{\partial}{\partial x} (x + y + z) = 2 \cdot \left(-\frac{1}{(1 + \varphi')^2} \right) \cdot \varphi'' \cdot \left(1 + \frac{\partial z}{\partial x} \right) \\ &= 2 \cdot \left(-\frac{1}{(1 + \varphi')^2} \right) \cdot \varphi'' \cdot \left(1 + \frac{2x - \varphi'}{1 + \varphi'} \right) = -\frac{2(2x + 1)\varphi''}{(1 + \varphi')^3} \end{aligned}$$

(注意, 式中 $\varphi' = \varphi'(x + y + z)$ 是由 $\varphi'(u)$ 与 $u = x + y + z$ 复合而成的函数, 而 z 又是 x, y 的函数)

14. 设函数 $z = z(x, y)$ 由方程 $F\left(x + \frac{z}{y}, y + \frac{z}{x}\right) = 0$ 所确定, 其中 F 可微, 且

$xF'_1 + yF'_2 \neq 0$, 证明 $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy$.

证 方法一、方程两边对 x 和 y 求偏导得

$$\begin{aligned} F'_1 \cdot \left(1 + \frac{1}{y} \frac{\partial z}{\partial x} \right) + F'_2 \cdot \frac{\frac{\partial z}{\partial x} \cdot x - z \cdot 1}{x^2} &= 0 \\ F'_1 \cdot \frac{\frac{\partial z}{\partial y} \cdot y - z \cdot 1}{y^2} + F'_2 \cdot \left(1 + \frac{1}{x} \frac{\partial z}{\partial y} \right) &= 0 \end{aligned}$$

解得

$$\frac{\partial z}{\partial x} = \frac{-yx^2F'_1 + yzF'_2}{x(xF'_1 + yF'_2)}, \quad \frac{\partial z}{\partial y} = \frac{xzF'_1 - xy^2F'_2}{y(xF'_1 + yF'_2)}$$

故

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{-yx^2F'_1 + yzF'_2}{xF'_1 + yF'_2} + \frac{xzF'_1 - xy^2F'_2}{xF'_1 + yF'_2} = z - xy$$

方法二、

设 $G(x, y, z) = F\left(x + \frac{z}{y}, y + \frac{z}{x}\right)$, 则 $G(x, y, z) = F\left(x + \frac{z}{y}, y + \frac{z}{x}\right) = 0$, 由

隐函数求导公式得

$$\begin{aligned} \frac{\partial z}{\partial x} &= -\frac{\frac{\partial G}{\partial x}}{\frac{\partial G}{\partial z}} = -\frac{F'_1 \cdot 1 + F'_2 \cdot \left(-\frac{z}{x^2}\right)}{F'_1 \cdot \frac{1}{y} + F'_2 \cdot \frac{1}{x}} = \frac{-yx^2F'_1 + yzF'_2}{x(xF'_1 + yF'_2)} \\ \frac{\partial z}{\partial y} &= -\frac{\frac{\partial G}{\partial y}}{\frac{\partial G}{\partial z}} = -\frac{F'_1 \cdot \left(-\frac{z}{y^2}\right) + F'_2 \cdot 1}{F'_1 \cdot \frac{1}{y} + F'_2 \cdot \frac{1}{x}} = \frac{xzF'_1 - xy^2F'_2}{y(xF'_1 + yF'_2)} \end{aligned}$$

余同.

方法三、方程两边取全微分得

$$\begin{aligned} F'_1 \cdot d\left(x + \frac{z}{y}\right) + F'_2 \cdot d\left(y + \frac{z}{x}\right) &= 0 \\ \Rightarrow F'_1 \cdot \left(dx + \frac{ydz - zdy}{y^2}\right) + F'_2 \cdot \left(dy + \frac{xdz - zdx}{x^2}\right) &= 0 \end{aligned}$$

解出 dz 得

$$dz = \frac{-yx^2F'_1 + yzF'_2}{xF'_1 + yF'_2} dx + \frac{xzF'_1 - xy^2F'_2}{xF'_1 + yF'_2} dy$$

所以

$$\frac{\partial z}{\partial x} = \frac{-yx^2F'_1 + yzF'_2}{x(xF'_1 + yF'_2)}, \quad \frac{\partial z}{\partial y} = \frac{xzF'_1 - xy^2F'_2}{y(xF'_1 + yF'_2)}$$

余同.

15. 设 $z = f(u)$, 而 $u = u(x, y)$ 由方程 $u = \varphi(u) + \int_y^x P(t)dt$ 所确定, 其中 f, φ 可微,

P 连续, 且 $\varphi' \neq 1$, 求 $P(y) \frac{\partial z}{\partial x} + P(x) \frac{\partial z}{\partial y}$.

解 方程 $u = \varphi(u) + \int_y^x P(t)dt$ 两边关于 x 和 y 求偏导得

$$\frac{\partial u}{\partial x} = \varphi'(u) \frac{\partial u}{\partial x} + P(x)$$

$$\frac{\partial u}{\partial y} = \varphi'(u) \frac{\partial u}{\partial y} - P(y)$$

解得

$$\frac{\partial u}{\partial x} = \frac{P(x)}{1 - \varphi'(u)}, \quad \frac{\partial u}{\partial y} = -\frac{P(y)}{1 - \varphi'(u)}$$

所以

$$\frac{\partial z}{\partial x} = f'(u) \frac{\partial u}{\partial x} = f'(u) \cdot \frac{P(x)}{1 - \varphi'(u)} = \frac{f'(u)P(x)}{1 - \varphi'(u)}$$

$$\frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y} = f'(u) \cdot \left(-\frac{P(y)}{1 - \varphi'(u)} \right) = -\frac{f'(u)P(y)}{1 - \varphi'(u)}$$

故

$$P(y) \frac{\partial z}{\partial x} + P(x) \frac{\partial z}{\partial y} = P(y) \cdot \frac{f'(u)P(x)}{1 - \varphi'(u)} + P(x) \cdot \left(-\frac{f'(u)P(y)}{1 - \varphi'(u)} \right) = 0$$

16. 设 $u = f(x, y, z)$ 有连续的二阶偏导数, 而 $z = z(x, y)$ 由方程 $xe^x - ye^y = ze^z$ 所确定, 求 du .

解 方程 $xe^x - ye^y = ze^z$ 关于 x 和 y 求偏导得

$$(x+1)e^x = (z+1)e^z \frac{\partial z}{\partial x}$$

$$-(y+1)e^y = (z+1)e^z \frac{\partial z}{\partial y}$$

解得

$$\frac{\partial z}{\partial x} = \frac{(x+1)e^x}{(z+1)e^z}, \quad \frac{\partial z}{\partial y} = -\frac{(y+1)e^y}{(z+1)e^z}$$

于是

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \frac{(x+1)e^x}{(z+1)e^z}$$

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} - \frac{\partial f}{\partial z} \frac{(y+1)e^y}{(z+1)e^z}$$

故

$$\begin{aligned} du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \\ &= \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \frac{(x+1)e^x}{(z+1)e^z} \right) dx + \left(\frac{\partial f}{\partial y} - \frac{\partial f}{\partial z} \frac{(y+1)e^y}{(z+1)e^z} \right) dy \end{aligned}$$

17. 设 $\begin{cases} 2x = v^2 - u^2 \\ y = uv \end{cases}$, 求 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}$.

解 共两个方程, 四个变量, 所以方程组确定两个二元函数, 由 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}$ 知, u 和 v 是函数,

所以 x 和 y 是自变量, 故 u 和 v 是 x 和 y 的二元函数 $u = u(x, y), v = v(x, y)$.

方法一、方程组关于 x 求偏导得

$$\begin{cases} 2 = 2v \frac{\partial v}{\partial x} - 2u \frac{\partial u}{\partial x} \\ 0 = \frac{\partial u}{\partial x} v + u \frac{\partial v}{\partial x} \end{cases}$$

解得

$$\frac{\partial u}{\partial x} = -\frac{u}{u^2 + v^2}, \frac{\partial v}{\partial x} = \frac{v}{u^2 + v^2}$$

方法二、设

$$F(x, y, u, v) = v^2 - u^2 - 2x, G(x, y, u, v) = uv - y$$

则

$$\begin{aligned} \frac{\partial u}{\partial x} &= -\frac{\frac{\partial(F, G)}{\partial(x, v)}}{\frac{\partial(F, G)}{\partial(u, v)}} = -\frac{\begin{vmatrix} -2 & 2v \\ 0 & u \end{vmatrix}}{\begin{vmatrix} -2u & 2v \\ v & u \end{vmatrix}} = -\frac{-2u}{-2u^2 - 2v^2} = -\frac{u}{u^2 + v^2} \\ \frac{\partial v}{\partial x} &= -\frac{\frac{\partial(F, G)}{\partial(u, x)}}{\frac{\partial(F, G)}{\partial(u, v)}} = -\frac{\begin{vmatrix} -2u & -2 \\ v & 0 \end{vmatrix}}{\begin{vmatrix} -2u & 2v \\ v & u \end{vmatrix}} = -\frac{2v}{-2u^2 - 2v^2} = \frac{v}{u^2 + v^2} \end{aligned}$$

18. 设 $x = \frac{1}{u} + \frac{1}{v}, y = \frac{1}{u^3} + \frac{1}{v^3}, z = uv - e^x$, 求 $\frac{\partial z}{\partial y}, \frac{\partial z}{\partial v}$.

解 共三个方程, 五个变量, 所以方程组确定三个二元函数, 由 $\frac{\partial z}{\partial y}, \frac{\partial z}{\partial v}$ 知, y 和 v 是自变

量, 故 x, z, u 都是 y 和 v 的二元函数 $x = x(y, v), z = z(y, v), u = u(y, v)$.

方法一、方程组关于 y 求偏导得

$$\begin{cases} \frac{\partial x}{\partial y} = -\frac{1}{u^2} \frac{\partial u}{\partial y} \\ 1 = -3 \frac{1}{u^4} \frac{\partial u}{\partial y} \\ \frac{\partial z}{\partial y} = \frac{\partial u}{\partial y} v - e^x \frac{\partial x}{\partial y} \end{cases}$$

解出 $\frac{\partial z}{\partial y}$ 得

$$\frac{\partial z}{\partial y} = \left(-\frac{1}{3} u^4\right) v - e^x \left(\frac{1}{3} u^2\right) = -\frac{1}{3} u^4 v - \frac{1}{3} u^2 e^x$$

方程组关于 v 求偏导得

$$\begin{cases} \frac{\partial x}{\partial v} = -\frac{1}{u^2} \frac{\partial u}{\partial v} - \frac{1}{v^2} \\ 0 = -3 \frac{1}{u^4} \frac{\partial u}{\partial v} - 3 \frac{1}{v^4} \\ \frac{\partial z}{\partial v} = \frac{\partial u}{\partial v} v + u - e^x \frac{\partial x}{\partial v} \end{cases}$$

解出 $\frac{\partial z}{\partial v}$ 得

$$\frac{\partial z}{\partial v} = \left(-\frac{u^4}{v^4}\right) v + u - e^x \left(\frac{u^2}{v^4} - \frac{1}{v^2}\right) = -\frac{u^4}{v^3} + u - \frac{u^2 e^x}{v^4} + \frac{e^x}{v^2}$$

方法二、设

$$F(x, y, z, u, v) = \frac{1}{u} + \frac{1}{v} - x, G(x, y, z, u, v) = \frac{1}{u^3} + \frac{1}{v^3} - y,$$

$$H(x, y, z, u, v) = uv - e^x - z$$

则

$$\frac{\partial z}{\partial y} = - \frac{\frac{\partial(F, G, H)}{\partial(x, y, u)}}{\frac{\partial(F, G, H)}{\partial(x, z, u)}} = - \frac{\begin{vmatrix} -1 & 0 & -\frac{1}{u^2} \\ 0 & -1 & -3\frac{1}{u^4} \\ -e^x & 0 & v \end{vmatrix}}{\begin{vmatrix} -1 & 0 & -\frac{1}{u^2} \\ 0 & 0 & -3\frac{1}{u^4} \\ -e^x & -1 & v \end{vmatrix}}$$

$$= - \frac{v + \frac{e^x}{u^2}}{\frac{3}{u^4}} = -\frac{1}{3} u^4 v - \frac{1}{3} u^2 e^x$$

$$\frac{\partial z}{\partial y} = - \frac{\frac{\partial(F, G, H)}{\partial(x, v, u)}}{\frac{\partial(F, G, H)}{\partial(x, z, u)}} = - \frac{\begin{vmatrix} -1 & -\frac{1}{v^2} & -\frac{1}{u^2} \\ 0 & -3\frac{1}{v^4} & -3\frac{1}{u^4} \\ -e^x & u & v \end{vmatrix}}{\begin{vmatrix} -1 & 0 & -\frac{1}{u^2} \\ 0 & 0 & -3\frac{1}{u^4} \\ -e^x & -1 & v \end{vmatrix}}$$

$$= - \frac{\frac{3}{v^3} - \frac{3}{u^3} - \frac{3e^x}{u^4 v^2} + \frac{3e^x}{u^2 v^4}}{\frac{3}{u^4}} = -\frac{u^4}{v^3} + u - \frac{u^2 e^x}{v^4} + \frac{e^x}{v^2}$$

19. 设 $z = f(x, y, z)$, $y = \varphi(x, t)$, 其中 f, φ 具有连续的偏导数, 且

$$\frac{\partial f}{\partial z} \neq 1, \frac{\partial \varphi}{\partial x} \neq 0, \text{ 求 } \frac{\partial z}{\partial y}, \frac{\partial z}{\partial t}.$$

解 两个方程, 四个变量, 所以方程组 $z = f(x, y, z)$, $y = \varphi(x, t)$ 确定两个二元函数,

由 $\frac{\partial z}{\partial y}, \frac{\partial z}{\partial t}$ 知, y 和 t 是自变量, 所以 x 和 z 都是 y 和 t 的二元函数

$$x = x(y, t), z = z(y, t).$$

对方程组 $z = f(x, y, z), y = \varphi(x, t)$ 关于 y 求偏导得

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial y} \\ 1 &= \frac{\partial \varphi}{\partial x} \frac{\partial x}{\partial y}\end{aligned}$$

将 $\frac{\partial x}{\partial y}$ 消去并解出 $\frac{\partial z}{\partial y}$ 得

$$\frac{\partial z}{\partial y} = \frac{\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x}}{\frac{\partial \varphi}{\partial x} \left(1 - \frac{\partial f}{\partial z}\right)}$$

对方程组 $z = f(x, y, z), y = \varphi(x, t)$ 关于 t 求偏导得

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial t} \\ 0 &= \frac{\partial \varphi}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial \varphi}{\partial t}\end{aligned}$$

将 $\frac{\partial x}{\partial t}$ 消去并解出 $\frac{\partial z}{\partial t}$ 得

$$\frac{\partial z}{\partial t} = -\frac{\frac{\partial f}{\partial x} \frac{\partial \varphi}{\partial t}}{\frac{\partial \varphi}{\partial x} \left(1 - \frac{\partial f}{\partial z}\right)}$$

20. 设 $u = f(x, y, z), \varphi(x^2, e^y, z) = 0, y = \sin x$, 其中 f, φ 具有一阶连续偏导数,

且 $\frac{\partial \varphi}{\partial z} \neq 0$, 求 $\frac{du}{dx}$.

解 $\frac{dy}{dx} = \cos x$

由于 $y = \sin x$, 所以方程 $\varphi(x^2, e^y, z) = 0$ 确定一元函数 $z = z(x)$, 对

$\varphi(x^2, e^y, z) = 0$ 关于 x 求导得

$$\varphi_1' \cdot 2x + \varphi_2' \cdot e^y \cos x + \varphi_3' \cdot \frac{dz}{dx} = 0$$

解得

$$\frac{dz}{dx} = -\frac{2x\varphi_1' + e^y \cos x \varphi_2'}{\varphi_3'}$$

于是

$$\begin{aligned} \frac{du}{dx} &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cos x + \frac{\partial f}{\partial z} \frac{dz}{dx} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cos x - \frac{2x\varphi_1' + e^y \cos x \varphi_2'}{\varphi_3'} \frac{\partial f}{\partial z} \end{aligned}$$

21. 设 $z = f(x, u, v)$, $u = g(x, y)$, $v = h(x, y, u)$, 其中 f, g, h 均可微, 求

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}.$$

解 方法一、由于 $u = g(x, y)$, 所以方程 $v = h(x, y, u)$ 确定函数 $v = v(x, y)$, 对方程

组 $u = g(x, y)$, $v = h(x, y, u)$ 关于 x 和 y 求偏导得

$$\frac{\partial u}{\partial x} = \frac{\partial g}{\partial x}, \frac{\partial v}{\partial x} = \frac{\partial h}{\partial x} + \frac{\partial h}{\partial u} \frac{\partial g}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial g}{\partial y}, \frac{\partial v}{\partial y} = \frac{\partial h}{\partial y} + \frac{\partial h}{\partial u} \frac{\partial g}{\partial y}$$

所以

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \frac{\partial g}{\partial x} + \frac{\partial f}{\partial v} \left(\frac{\partial h}{\partial x} + \frac{\partial h}{\partial u} \frac{\partial g}{\partial x} \right) \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \frac{\partial g}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial u} \frac{\partial g}{\partial x} \\ \frac{\partial z}{\partial y} &= \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} \\ &= \frac{\partial f}{\partial u} \frac{\partial g}{\partial y} + \frac{\partial f}{\partial v} \left(\frac{\partial h}{\partial y} + \frac{\partial h}{\partial u} \frac{\partial g}{\partial y} \right) \\ &= \frac{\partial f}{\partial u} \frac{\partial g}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial u} \frac{\partial g}{\partial y} \end{aligned}$$

方法二、设

$$F(x, y, z, u, v) = z - f(x, u, v), G(x, y, z, u, v) = u - g(x, y),$$

$$H(x, y, z, u, v) = v - h(x, y, u)$$

$$\text{方程组} \begin{cases} F(x, y, z, u, v) = z - f(x, u, v) = 0 \\ G(x, y, z, u, v) = u - g(x, y) = 0 \\ H(x, y, z, u, v) = v - h(x, y, u) = 0 \end{cases} \quad \text{有三个方程，五个变量，所以该方程组确}$$

定三个二元函数，由 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ 知， x 和 y 为自变量，所以 z, u, v 都是自变量 x 和 y 的二元

函数 $z = z(x, y), u = g(x, y), v = v(x, y)$ ，由隐函数求导公式得

$$\begin{aligned} \frac{\partial z}{\partial x} &= - \frac{\frac{\partial(F, G, H)}{\partial(x, u, v)}}{\frac{\partial(F, G, H)}{\partial(z, u, v)}} = - \frac{\begin{vmatrix} -\frac{\partial f}{\partial x} & -\frac{\partial f}{\partial u} & -\frac{\partial f}{\partial v} \\ -\frac{\partial g}{\partial x} & 1 & 0 \\ -\frac{\partial h}{\partial x} & \frac{\partial h}{\partial u} & 1 \end{vmatrix}}{\begin{vmatrix} 1 & -\frac{\partial f}{\partial u} & -\frac{\partial f}{\partial v} \\ 0 & 1 & 0 \\ 0 & \frac{\partial h}{\partial u} & 1 \end{vmatrix}} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \frac{\partial g}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial u} \frac{\partial g}{\partial x} \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= - \frac{\frac{\partial(F, G, H)}{\partial(y, u, v)}}{\frac{\partial(F, G, H)}{\partial(z, u, v)}} = - \frac{\begin{vmatrix} 0 & -\frac{\partial f}{\partial u} & -\frac{\partial f}{\partial v} \\ -\frac{\partial g}{\partial y} & 1 & 0 \\ -\frac{\partial g}{\partial y} & \frac{\partial h}{\partial u} & 1 \end{vmatrix}}{\begin{vmatrix} 1 & -\frac{\partial f}{\partial u} & -\frac{\partial f}{\partial v} \\ 0 & 1 & 0 \\ 0 & \frac{\partial h}{\partial u} & 1 \end{vmatrix}} \\ &= \frac{\partial f}{\partial u} \frac{\partial g}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial h}{\partial u} \frac{\partial g}{\partial y} \end{aligned}$$

22. 设函数 $f(u, v)$ 具有连续的二阶偏导数, 且满足方程 $f'_u(u, v) + f'_v(u, v) = uv$, 求函数

$y = e^{-2x} f(x, x)$ 所满足的一阶微分方程, 并求其通解.

解 对 $y = e^{-2x} f(x, x)$ 求导得

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [e^{-2x} f(x, x)] = -2e^{-2x} f(x, x) + e^{-2x} \frac{d}{dx} f(x, x) \\ &= -2e^{-2x} f(x, x) + e^{-2x} \left[f'_u(x, x) \cdot \frac{dx}{dx} + f'_v(x, x) \cdot \frac{dx}{dx} \right] \\ &= -2e^{-2x} f(x, x) + e^{-2x} [f'_u(x, x) + f'_v(x, x)] = -2e^{-2x} f(x, x) + e^{-2x} x^2 \\ &= -2y + x^2 e^{-2x}\end{aligned}$$

(注意, $f(x, x)$ 由 $f(u, v)$ 与 $u = x, v = x$ 复合而成, 此外, 在 $f'_u(u, v) + f'_v(u, v) = uv$

中令 $u = x, v = x$ 得 $f'_u(x, x) + f'_v(x, x) = x^2$)

所以所求一阶微分方程为

$$\frac{dy}{dx} + 2y = x^2 e^{-2x}$$

为一阶线性微分方程, 其通解为

$$y = e^{-\int 2dx} \left(\int x^2 e^{-2x} e^{\int 2dx} dx + C \right) = e^{-2x} \left(\int x^2 dx + C \right) = e^{-2x} \left(\frac{x^3}{3} + C \right)$$

即

$$y = e^{-2x} \left(\frac{x^3}{3} + C \right)$$

23. 设函数 $f(u)$ 具有连续的二阶导数, 且 $f(0) = 0, f'(0) = 0$, 又函数 $z = f(e^x \cos y)$

满足方程 $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (4z + e^x \cos y) e^{2x}$, 求函数 $f(u)$ 的表达式.

解 令 $u = e^x \cos y$, 则 $z = f(u)$, 求偏导得

$$\frac{\partial z}{\partial x} = f'(u) \frac{\partial u}{\partial x} = f'(u) e^x \cos y$$

$$\frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y} = f'(u) (-e^x \sin y)$$

求二阶偏导数得

$$\begin{aligned}
\frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} (f'(u)e^x \cos y) = \frac{\partial}{\partial x} (f'(u)) \cdot e^x \cos y + f'(u)e^x \cos y \\
&= \left(f''(u) \frac{\partial u}{\partial x} \right) e^x \cos y + f'(u)e^x \cos y = f''(u)(e^x \cos y)^2 + f'(u)e^x \cos y \\
\frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y} [f'(u)(-e^x \sin y)] = \frac{\partial}{\partial y} (f'(u)) \cdot (-e^x \sin y) + f'(u)(-e^x \cos y) \\
&= \left(f''(u) \frac{\partial u}{\partial y} \right) \cdot (-e^x \sin y) + f'(u)(-e^x \cos y) = f''(u)(-e^x \sin y)^2 + f'(u)(-e^x \cos y)
\end{aligned}$$

所以

$$\begin{aligned}
\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} &= \left[f''(u)(e^x \cos y)^2 + f'(u)e^x \cos y \right] \\
&\quad + \left[f''(u)(-e^x \sin y)^2 + f'(u)(-e^x \cos y) \right] \\
&= f''(u)e^{2x} = (4z + e^x \cos y)e^{2x} = (4f(u) + u)e^{2x}
\end{aligned}$$

故

$$f''(u) - 4f(u) = u$$

其通解为

$$f(u) = C_1 e^{2u} + C_2 e^{-2u} - \frac{1}{4} u$$

由 $f(0) = 0, f'(0) = 0$ 得 $C_1 = \frac{1}{16}, C_2 = -\frac{1}{16}$, 于是

$$f(u) = \frac{1}{16} e^{2u} - \frac{1}{16} e^{-2u} - \frac{1}{4} u$$

24. 设函数 $z = f(x, y)$ 具有连续的二阶偏导数, 且满足方程

$$4 \frac{\partial^2 z}{\partial x^2} + 12 \frac{\partial^2 z}{\partial x \partial y} + 5 \frac{\partial^2 z}{\partial y^2} = 0, \text{ 确定 } a, b \text{ 的值, 使方程在变换 } \begin{cases} \xi = x + ay \\ \eta = x + by \end{cases} \text{ 下简化}$$

$$\text{为 } \frac{\partial^2 z}{\partial \xi \partial \eta} = 0.$$

解 求偏导得

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial z}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial z}{\partial \xi} + \frac{\partial z}{\partial \eta}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial z}{\partial \eta} \frac{\partial \eta}{\partial y} = a \frac{\partial z}{\partial \xi} + b \frac{\partial z}{\partial \eta}$$

求二阶偏导数得

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial \xi} + \frac{\partial z}{\partial \eta} \right) \\ &= \left(\frac{\partial^2 z}{\partial \xi^2} \frac{\partial \xi}{\partial x} + \frac{\partial^2 z}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial x} \right) + \left(\frac{\partial^2 z}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial^2 z}{\partial \eta^2} \frac{\partial \eta}{\partial x} \right) \\ &= \left(\frac{\partial^2 z}{\partial \xi^2} + \frac{\partial^2 z}{\partial \xi \partial \eta} \right) + \left(\frac{\partial^2 z}{\partial \eta \partial \xi} + \frac{\partial^2 z}{\partial \eta^2} \right) = \frac{\partial^2 z}{\partial \xi^2} + 2 \frac{\partial^2 z}{\partial \xi \partial \eta} + \frac{\partial^2 z}{\partial \eta^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial \xi} + \frac{\partial z}{\partial \eta} \right) \\ &= \left(\frac{\partial^2 z}{\partial \xi^2} \frac{\partial \xi}{\partial y} + \frac{\partial^2 z}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial y} \right) + \left(\frac{\partial^2 z}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial^2 z}{\partial \eta^2} \frac{\partial \eta}{\partial y} \right) \\ &= \left(a \frac{\partial^2 z}{\partial \xi^2} + b \frac{\partial^2 z}{\partial \xi \partial \eta} \right) + \left(a \frac{\partial^2 z}{\partial \eta \partial \xi} + b \frac{\partial^2 z}{\partial \eta^2} \right) = a \frac{\partial^2 z}{\partial \xi^2} + (a + b) \frac{\partial^2 z}{\partial \xi \partial \eta} + b \frac{\partial^2 z}{\partial \eta^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y} \left(a \frac{\partial z}{\partial \xi} + b \frac{\partial z}{\partial \eta} \right) \\ &= a \left(\frac{\partial^2 z}{\partial \xi^2} \frac{\partial \xi}{\partial y} + \frac{\partial^2 z}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial y} \right) + b \left(\frac{\partial^2 z}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial^2 z}{\partial \eta^2} \frac{\partial \eta}{\partial y} \right) \\ &= a \left(a \frac{\partial^2 z}{\partial \xi^2} + b \frac{\partial^2 z}{\partial \xi \partial \eta} \right) + b \left(a \frac{\partial^2 z}{\partial \eta \partial \xi} + b \frac{\partial^2 z}{\partial \eta^2} \right) = a^2 \frac{\partial^2 z}{\partial \xi^2} + 2ab \frac{\partial^2 z}{\partial \xi \partial \eta} + b^2 \frac{\partial^2 z}{\partial \eta^2}\end{aligned}$$

代入方程得

$$\begin{aligned}& (5a^2 + 12a + 4) \frac{\partial^2 z}{\partial \xi^2} + (10ab + 12a + 12b + 8) \frac{\partial^2 z}{\partial \xi \partial \eta} \\ & + (5b^2 + 12b + 4) \frac{\partial^2 z}{\partial \eta^2} = 0\end{aligned}$$

所以

$$\begin{cases} 5a^2 + 12a + 4 = 0 \\ 5b^2 + 12b + 4 = 0 \end{cases}$$

解得

$$a = -2, a = -\frac{2}{5}, b = -2, b = -\frac{2}{5}$$

为使 $10ab + 12a + 12b + 8 \neq 0$ ，故应取 $a = -2, b = -\frac{2}{5}$ 或 $a = -\frac{2}{5}, b = -2$ 。

25. 设 $u = x + y, v = x - y, w = xy - z$ ，试将方程 $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ 变

换为 w 关于自变量 u, v 的偏导数的方程, 并求原方程的解.

解 由 $w = xy - z$ 得

$$z = xy - w$$

求偏导得

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (xy - w) = y - \left(\frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x} \right) = y - \left(\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} \right)$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (xy - w) = x - \left(\frac{\partial w}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial y} \right) = x - \left(\frac{\partial w}{\partial u} - \frac{\partial w}{\partial v} \right)$$

求二阶偏导数得

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left[y - \left(\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} \right) \right] = 0 - \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial u} \right) - \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial v} \right) \\ &= - \left[\frac{\partial^2 w}{\partial u^2} \frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial u \partial v} \frac{\partial v}{\partial x} \right] - \left[\frac{\partial^2 w}{\partial v \partial u} \frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial x} \right] \\ &= - \left[\frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial u \partial v} \right] - \left[\frac{\partial^2 w}{\partial v \partial u} + \frac{\partial^2 w}{\partial v^2} \right] = - \left(\frac{\partial^2 w}{\partial u^2} + 2 \frac{\partial^2 w}{\partial u \partial v} + \frac{\partial^2 w}{\partial v^2} \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} \left[y - \left(\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} \right) \right] = 1 - \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial u} \right) - \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial v} \right) \\ &= - \left[\frac{\partial^2 w}{\partial u^2} \frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial u \partial v} \frac{\partial v}{\partial y} \right] - \left[\frac{\partial^2 w}{\partial v \partial u} \frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial y} \right] \\ &= - \left[\frac{\partial^2 w}{\partial u^2} - \frac{\partial^2 w}{\partial u \partial v} \right] - \left[\frac{\partial^2 w}{\partial v \partial u} - \frac{\partial^2 w}{\partial v^2} \right] = - \left(\frac{\partial^2 w}{\partial u^2} - \frac{\partial^2 w}{\partial v^2} \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y} \left[x - \left(\frac{\partial w}{\partial u} - \frac{\partial w}{\partial v} \right) \right] = 0 - \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial u} \right) + \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial v} \right) \\ &= - \left[\frac{\partial^2 w}{\partial u^2} \frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial u \partial v} \frac{\partial v}{\partial y} \right] + \left[\frac{\partial^2 w}{\partial v \partial u} \frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial v^2} \frac{\partial v}{\partial y} \right] \\ &= - \left[\frac{\partial^2 w}{\partial u^2} - \frac{\partial^2 w}{\partial u \partial v} \right] + \left[\frac{\partial^2 w}{\partial v \partial u} - \frac{\partial^2 w}{\partial v^2} \right] = - \left(\frac{\partial^2 w}{\partial u^2} - 2 \frac{\partial^2 w}{\partial u \partial v} + \frac{\partial^2 w}{\partial v^2} \right) \end{aligned}$$

代入方程并整理得

$$\frac{\partial^2 w}{\partial u^2} = \frac{1}{2},$$

为所求方程.

$$\text{方程 } \frac{\partial^2 w}{\partial u^2} = \frac{1}{2} \text{ 对 } u \text{ 积分得}$$

$$\frac{\partial w}{\partial u} = \frac{1}{2}u + \varphi(v)$$

再对 u 积分得

$$w = \frac{1}{4}u^2 + \varphi(v)u + \psi(v)$$

其中 $\varphi(v), \psi(v)$ 是任意函数.

因此原方程的通解为

$$xy - z = \frac{1}{4}(x+y)^2 + (x+y)\varphi(x-y) + \psi(x-y)$$

26. 取 x 为函数, $u = y - z$ 和 $v = y + z$ 为自变量, 变换方程

$$(y - z) \frac{\partial z}{\partial x} + (y + z) \frac{\partial z}{\partial y} = 0.$$

解 由 $u = y - z$ 和 $v = y + z$ 得

$$y = \frac{u + v}{2}, z = \frac{v - u}{2}$$

由于 z 是 x 和 y 的函数 $z = z(x, y)$, 所以

$$\frac{v - u}{2} = z(x, \frac{u + v}{2})$$

其中 x 是 u 和 v 的函数 $x = x(u, v)$, 上式关于 u 和 v 求偏导得

$$-\frac{1}{2} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{1}{2}$$

$$\frac{1}{2} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{1}{2}$$

两式相加得

$$0 = \frac{\partial z}{\partial x} \left(\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v} \right) + \frac{\partial z}{\partial y}$$

由 $(y - z) \frac{\partial z}{\partial x} + (y + z) \frac{\partial z}{\partial y} = 0$ 得

$$\frac{\partial z}{\partial y} = -\frac{y - z}{y + z} \frac{\partial z}{\partial x} = -\frac{u}{v} \frac{\partial z}{\partial x}$$

代入前面的式子得

$$0 = \frac{\partial z}{\partial x} \left(\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v} \right) - \frac{u}{v} \frac{\partial z}{\partial x}$$

因此变换所得方程为

$$\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v} = \frac{u}{v}$$

27. 证明：若可微函数 $u = f(x, y, z)$ 满足方程 $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$ ，则

$f(x, y, z)$ 是 n 次齐次函数，即 $f(tx, ty, tz) = t^n f(x, y, z)$ 。

证 令 $\varphi(t) = f(tx, ty, tz)$ ，求导得

$$\begin{aligned}\varphi'(t) &= \frac{d}{dt} f(tx, ty, tz) \\&= f_1'(tx, ty, tz) \cdot x + f_2'(tx, ty, tz) \cdot y + f_3'(tx, ty, tz) \cdot z \\&= \frac{1}{t} [txf_1'(tx, ty, tz) + tyf_2'(tx, ty, tz) + tzf_3'(tx, ty, tz)]\end{aligned}$$

因为 $u = f(x, y, z)$ 满足方程 $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$ ，所以

$$xf_1'(x, y, z) + yf_2'(x, y, z) + zf_3'(x, y, z) = nf(x, y, z)$$

于是

$$txf_1'(tx, ty, tz) + tyf_2'(tx, ty, tz) + tzf_3'(tx, ty, tz) = nf(tx, ty, tz) = n\varphi(t)$$

从而

$$\varphi'(t) = \frac{1}{t} n\varphi(t)$$

为可分离变量方程，分离变量得

$$\frac{d\varphi(t)}{\varphi(t)} = n \frac{dt}{t}$$

积分得

$$\begin{aligned}\ln|\varphi(t)| &= n \ln|t| + C_1 \\ \Rightarrow \ln|\varphi(t)| &= n \ln|t| + \ln e^{C_1} = \ln|e^{C_1} t^n| \\ \Rightarrow \varphi(t) &= \pm e^{C_1} t^n = C t^n \quad (C = \pm e^{C_1})\end{aligned}$$

令 $t = 1$ 得 $C = \varphi(1) = f(x, y, z)$, 所以

$$\varphi(t) = t^n f(x, y, z)$$

故

$$f(tx, ty, tz) = t^n f(x, y, z)$$